Лабораторная работа №5

Расширенная настройка HTTP-сервера Apache

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Группа: НФИбд 02–23

дисциплина: **Администрирование сетевых подсистем (Lab 5)**

Цель работы

Приобретение практических навыков по расширенному конфигурированию HTTP-сервера Apache в части безопасности и возможности использования PHP.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

```
Far Manager, version 3.0.6364.0 x64
Copyright © 1996-2000 Eugene Roshal, Copyright © 2000-2024 Far Group

C:\work\claudely\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Resuming suspended VM...
==> server: Booting VM...
```

Рис. 1.1. Открытие рабочего каталога с проектом и запуск виртуальной машины server.

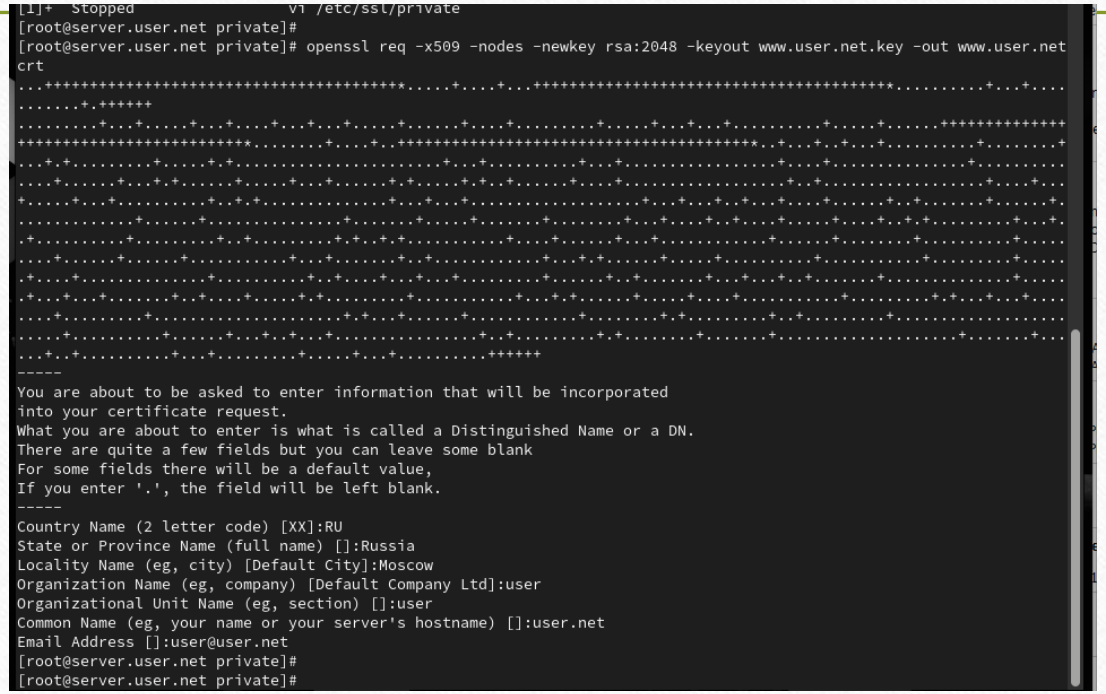
Конфигурирование HTTP-сервера для работы через протокол HTTPS

root@server:/etc/pki/tls/private

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# mkdir -p /etc/pki/tls/private
[root@server.user.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.user.net ~]# cd /etc/pki/tls/private
[root@server.user.net private]#
```

Рис. 1.2. Переход в режим суперпользователя и создание в каталоге /etc/ssl каталога private.

Конфигурирование HTTP-сервера для работы через протокол HTTPS



```
[1]+  Stopped                 vi /etc/ssl/private
[root@server.user.net private]#
[root@server.user.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.user.net.key -out www.user.net
crt
...++++++++++++++++++++++++++++++++++++++++++*.....+....+...+++++++++++++++++++++++++++++++++++++++*..........+...+....
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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:user
Organizational Unit Name (eg, section) []:user
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server.user.net private]#
[root@server.user.net private]#
```

Рис. 1.3. Генерация ключа.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

```
cert.pem  certs  ct_log_list.cnf  openssl.cnf  private
[root@server.claudely.net ssl]#
[root@server.claudely.net ssl]# cp /etc/ssl/private/www.claudely.net.crt /etc/ssl/certs
cp: overwrite '/etc/ssl/certs/www.claudely.net.crt'? yes
[root@server.claudely.net ssl]#
```

Рис. 1.4. Генерация сертификата.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

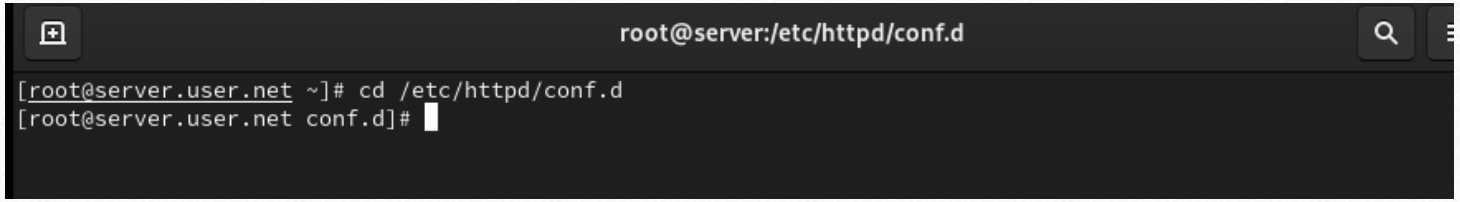


Рис. 1.5. Открытие каталога с конфигурационными файлами.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

root@server:/etc/httpd/conf.d

```
<VirtualHost *:80>
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  RewriteEngine on
  RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
 </VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
  SSLEngine on
  ServerAdmin webmaster@user.net
  DocumentRoot /var/www/html/www.user.net
  ServerName www.user.net
  ServerAlias www.user.net
  ErrorLog logs/www.user.net-error_log
  CustomLog logs/www.user.net-access_log common
  SSLCertificateFile /etc/ssl/certs/www.user.net.crt
  SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
~
~
```

Рис. 1.6. Открытие файла /etc/httpd/conf.d/www.user .net.conf на редактирование и замена содержимого.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

```
[root@server.user.net ~]#
[root@server.user.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps mountd nfs ntp pop3 pop3s rpc-bind samba smtp smtp-submission s
sh ssh-custom
[root@server.user.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2 bacula
 bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-t
estnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector crate
db ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
 dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-l
daps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availabili
ty http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect
kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube
-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client
llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur
mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storagecon
sole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-e
xporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rq
uotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom steam-streaming svdrp svn syncthi
ng syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-c
lient upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disc
overy-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-serve
r zerotier
[root@server.user.net ~]# firewall-cmd --add-service=https
Warning: ALREADY_ENABLED: 'https' already in 'public'
success
[root@server.user.net ~]# firewall-cmd --add-service=https --permanent
Warning: ALREADY_ENABLED: https
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

Рис. 1.7. Внесение изменений в настройки межсетевого экрана на сервере, разрешив работу с https.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

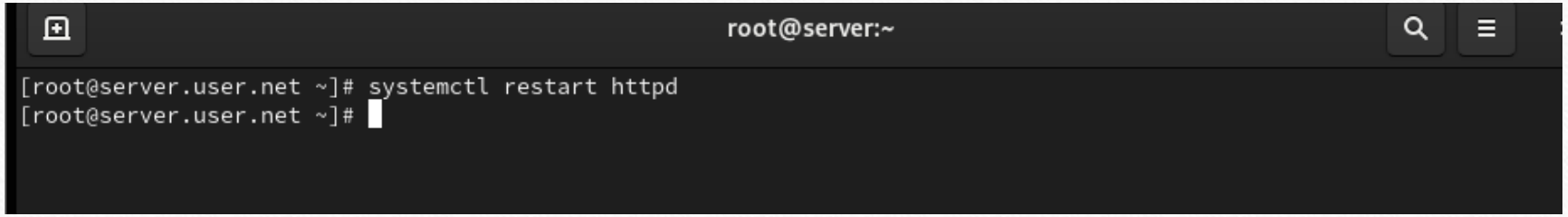


Рис. 1.8. Перезапуск веб-сервера.

Конфигурирование HTTP-сервера для работы через протокол HTTPS

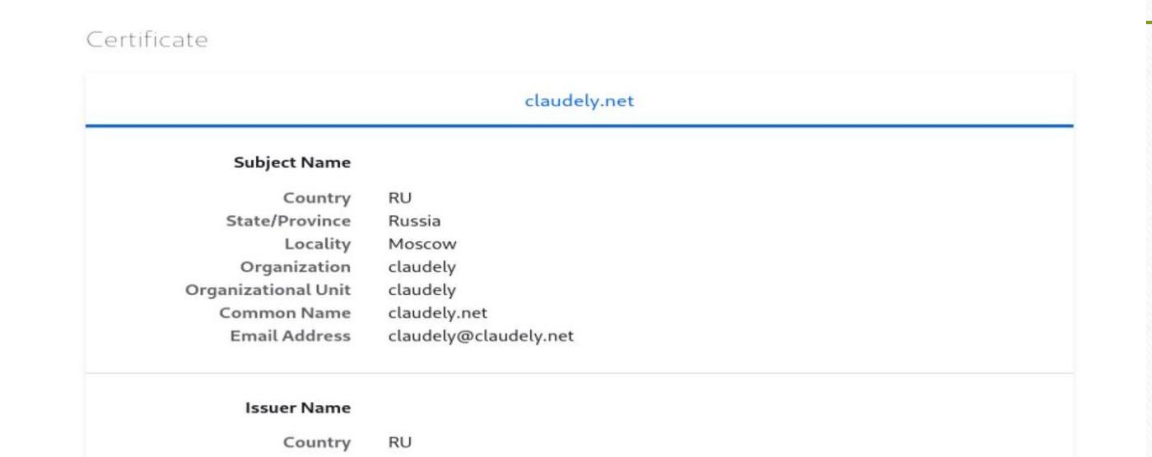


Рис. 1.9. Открытие веб-сервера www.claudely.net и автоматическое переключение на работу по протоколу HTTPS.

Конфигурирование HTTP-сервера для работы с PHP

```
[root@server.user.net ~]# dnf -y install php
Last metadata expiration check: 0:15:28 ago on Fri 06 Mar 2026 07:45:45 PM UTC.
Dependencies resolved.
================================================================================================================
 Package                        Architecture        Version                        Repository              Size
================================================================================================================
Installing:
 php                            x86_64              8.0.30-5.el9_7                 appstream              8.8 k
Installing dependencies:
 nginx-filesystem               noarch              2:1.20.1-24.el9                appstream              9.2 k
 php-common                     x86_64              8.0.30-5.el9_7                 appstream              667 k
Installing weak dependencies:
 php-cli                        x86_64              8.0.30-5.el9_7                 appstream              3.1 M
 php-fpm                        x86_64              8.0.30-5.el9_7                 appstream              1.6 M
 php-mbstring                   x86_64              8.0.30-5.el9_7                 appstream              468 k
 php-opcache                    x86_64              8.0.30-5.el9_7                 appstream              510 k
 php-pdo                        x86_64              8.0.30-5.el9_7                 appstream               82 k
 php-xml                        x86_64              8.0.30-5.el9_7                 appstream              133 k

Transaction Summary
================================================================================================================
Install  9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginx-filesystem-1.20.1-24.el9.noarch.rpm                                 36 kB/s | 9.2 kB     00:00
(2/9): php-8.0.30-5.el9_7.x86_64.rpm                                             28 kB/s | 8.8 kB     00:00
(3/9): php-common-8.0.30-5.el9_7.x86_64.rpm                                     188 kB/s | 667 kB     00:03
(4/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm                                        315 kB/s | 1.6 MB     00:05
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm                                   133 kB/s | 468 kB     00:03
(6/9): php-opcache-8.0.30-5.el9_7.x86_64.rpm                                    208 kB/s | 510 kB     00:02
(7/9): php-pdo-8.0.30-5.el9_7.x86_64.rpm                                         67 kB/s |  82 kB     00:01
(8/9): php-xml-8.0.30-5.el9_7.x86_64.rpm                                        190 kB/s | 133 kB     00:00
(9/9): php-cli-8.0.30-5.el9_7.x86_64.rpm                                        192 kB/s | 3.1 MB     00:16
----------------------------------------------------------------------------------------------------------------
Total                                                                           388 kB/s | 6.5 MB     00:17
```

Рис. 2.1. Установка пакетов для работы с PHP.

Конфигурирование HTTP-сервера для работы с PHP

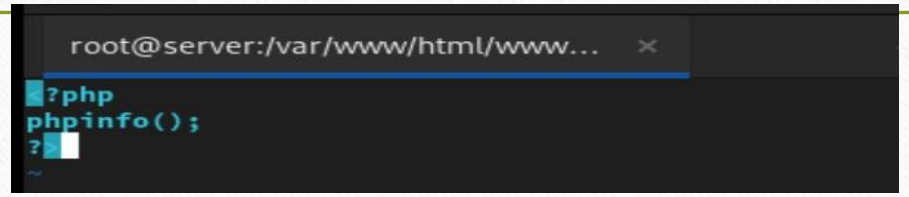


Рис. 2.2. Замена файла index.html на index.php с содержанием из лабораторной работы.

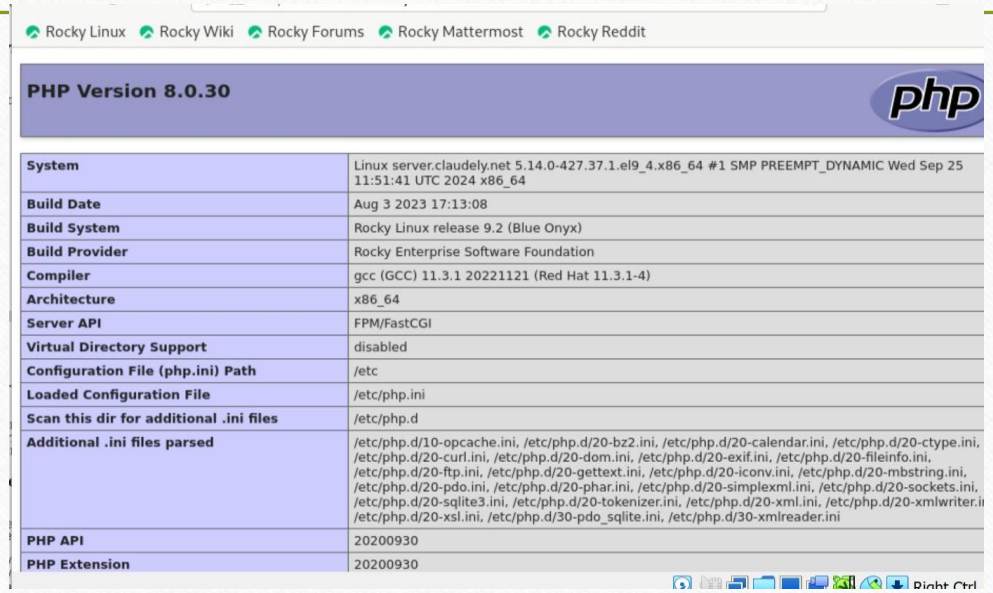
Конфигурирование HTTP-сервера для работы с PHP

```
[root@server.user.net ~]# chown -R apache:apache /var/www
[root@server.user.net ~]# restorecon -vR /etc
Relabeled /etc/dovecot from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dovecot_etc_t:s0
Relabeled /etc/dovecot/conf.d from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dovecot_etc_t:s0
Relabeled /etc/dovecot/conf.d/auth-system.conf.ext from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dove
cot_etc_t:s0
Relabeled /etc/dovecot/conf.d/10-master.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dovecot_et
c_t:s0
Relabeled /etc/dovecot/conf.d/.10-mail.conf.swp from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dovecot
_etc_t:s0
Relabeled /etc/dovecot/dovecot.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:dovecot_etc_t:s0
Relabeled /etc/dovecot.bak from system_u:object_r:dovecot_etc_t:s0 to system_u:object_r:etc_t:s0
Relabeled /etc/dovecot.bak/conf.d from system_u:object_r:dovecot_etc_t:s0 to system_u:object_r:etc_t:s0
Relabeled /etc/dovecot.bak/conf.d/10-auth.conf from system_u:object_r:dovecot_etc_t:s0 to system_u:object_r:etc_t:s0
Relabeled /etc/dovecot.bak/conf.d/10-director.conf from system_u:object_r:dovecot_etc_t:s0 to system_u:object_r:etc_
```

```
[root@server.user.net ~]# restorecon -vR /var/www
[root@server.user.net ~]# systemctl restart httpd
[root@server.user.net ~]#
```

Рис. 2.3. Корректировка прав доступа в каталог с веб-контентом, восстановление контекста безопасности в SELinux и перезапуск HTTP-сервера.

Конфигурирование HTTP-сервера для работы с PHP



Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

PHP Version 8.0.30

System	Linux server.claudely.net 5.14.0-427.37.1.el9_4.x86_64 #1 SMP PREEMPT_DYNAMIC Wed Sep 25 11:51:41 UTC 2024 x86_64
Build Date	Aug 3 2023 17:13:08
Build System	Rocky Linux release 9.2 (Blue Onyx)
Build Provider	Rocky Enterprise Software Foundation
Compiler	gcc (GCC) 11.3.1 20221121 (Red Hat 11.3.1-4)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.i /etc/php.d/20-xsl.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini
PHP API	20200930
PHP Extension	20200930

Рис. 2.4. Проверка вывода страницы с информацией об используемой на веб-сервере версии PHP.

Внесение изменений в настройки внутреннего окружения виртуальной машины

```
[root@server.user.net ~]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
[root@server.user.net ~]#
[root@server.user.net ~]#
[root@server.user.net ~]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/www.user.net/index.html'?
[root@server.user.net ~]#
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.user.net ~]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.user.net ~]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pki/tls/pri
vate
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
cp: cannot stat '/etc/pki/tls/certs/www.user.net.crt': No such file or directory
[root@server.user.net ~]# cp -R /etc/pki/tls/certs/www.user.net.crt  /vagrant/provision/server/http/etc/pki/tls/cert
s
```

Рис. 3.1. Внесение изменений в настройки внутреннего окружения /vagrant/provision/server/http и копирование конфигурационных файлов в каталоги.

Внесение изменений в настройки внутреннего окружения виртуальной машины

root@server:/vagrant/provision/server/http

```
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
dnf -y install php

echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
cp -R /vagrant/provision/server/http/etc/tls/* /etc/tls

chown -R apache:apache /var/www

restorecon -vR /etc
restorecon -vR /var/www

echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent

echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

Рис. 3.2. Внесение изменений в скрипт /vagrant/provision/server/http.sh, добавив установку PHP и настройку межсетевого экрана, разрешающую работать с https.

Вывод

В ходе выполнения лабораторной работы были приобретены практические навыки по расширенному конфигурированию HTTPсервера Apache в части безопасности и возможности использования PHP.

Спасибо за внимание!